CITY
ST GEORGE'S
UNIVERSITY OF LONDON

School of Science & Technology

IN1011 Operating Systems
Stage 1 Mock Examination A

May/June

Examination duration: **90** minutes

Division of marks: Answer **3** of the **4** questions (20 marks for each question). If more than **3** questions are answered, the best **3** will be counted. Total mark is out of **60**

Other instructions: All necessary working must be shown.

BEGIN EACH QUESTION ON A FRESH PAGE

Number of answer books to be provided: 1
Calculators permitted: Yes
Dictionaries permitted: None
Can question paper be removed from the examination room: No

Additional materials: **None**

Question 1

a. Program A() is executed twice on a system. The executions are not concurrent. The global variable *flag* is initialised to **true** for one of the executions, and **false** for the other execution.

```
A( ){
    while(flag){
        int n = 1;
        flag = false;
    }
}
```

i. Each execution will have an associated process image. Give 2 similarities and 2 differences between the process images. **[4 Marks]**

ii. What does it mean for a process to be *CPU-bound*? Is A() CPU-bound? Why? **[3 Marks]**

b. The OS uses control stacks when interrupts occur.

i. What is the purpose of a control stack when the OS handles an interrupt? Your answer should include 2 examples of data copied to the stack. **[3 Marks]**

ii. What property shared by all stacks makes a control stack very useful to the OS when nested interrupts occur? Your answer should name the property and briefly explain why this property is useful for nested routines. **[3 Marks]**

c. A common technique for achieving mutual exclusion on a uniprocessor system is disabling interrupts before entering a critical section and enabling interrupts after exiting a critical section.

i. Why does this work? **[2 Marks]**

ii. Why won't this work on a multiprocessor system? **[2 Marks]**

d. A typical hardware architecture provides an instruction called *return from interrupt* (RETI). This instruction switches the CPU mode of operation from kernel to user mode.

i. Which OS routines would invoke this instruction? **[1 Mark]**

ii. Speculate on what happens if a user program invokes this instruction. **[2 Marks]**

Question 2

a. Upon creation, 3 processes (A, B and C) are admitted into the ready queue in quick succession – i.e., one immediately after another. You notice the processes always successfully terminate in the order A, B, C – and the average turnaround time is *always the same* – irrespective of the order in which the processes are first admitted into the ready queue. The processes make no IO requests and have different service times.

   i. What scheduling policy could the OS be employing? Explain why you think this. **[3 Marks]**

   ii. Name 2 scheduling policies that are not employed by the OS. Explain why you think this for each policy you mention. **[4 Marks]**

b. Suppose the 3 processes in question 3b always successfully terminate in the order that they are first admitted to the ready queue. And, that the average turnaround time *is dependent* on the order of first admittance to the ready queue.

   i. What scheduling policy could the OS be employing? Explain why you think this. **[3 Marks]**

   ii. Name 3 scheduling policies that are not employed by the OS. Explain why you think this for each policy you mention. **[6 Marks]**

c. Finally, suppose the 3 processes in question 3b always terminate in the reverse order in which they are first admitted to the ready queue (you may ignore turnaround times). Briefly describe a scheduling policy that ensures this happens. Your description should clearly state the *selection function* and *decision mode* for the policy. **[4 Marks]**

Question 3

a. In the context of demand-paging, what does TLB stand for? Explain why the TLB is necessary to achieve acceptable performance.

[5 Marks]

b. Give a formula for EAT in terms of ***α*** (the base access time), ***p*** (the page-fault rate) and ***S*** (the average page-fault service time).

[4 Marks]

c. In a virtual memory system based on paging, what is a page table? Draw a diagram to show how a page table (without any translation look-aside buffer) is used to translate a virtual memory address to a physical memory address.

[6 Marks]

d. Calculate the Effective Access Time and Slowdown rate with the following conditions:
   i. base memory access time = 200ns and average page-fault service time = 0.1ms.
   ii. suppose one memory reference in every 10000 results in a page-fault.

[5 Marks]

Question 3

a. Consider a swapping system that has a series of holes (all in MB) with the following sizes (in order): **9, 20, 15, 22, 18, 17, 19, 31, 11, 14**

Three processes of sizes **9, 11, 13** MB arrive in order. Select the correct order of the holes selected if a best fit strategy is used.

[5 Marks]

b. Consider the following scenario. There are two users, user one is called troy and belongs to a group called `troy`. The second user is called `zapata` and is NOT in the group `troy`. Then, user troy runs two ls commands in his home directory `/home/troy`, running a bash shell, with the following results:

```
troy@debian:~$ ls –al
total 12
drwxr-xr-x  3 troy troy 4096 Mar 20 22:12 .
drwxr-xr-x 22 troy troy 4096 Mar 20 22:12 ..
drwxr-x--x  2 troy troy 4096 Mar 20 21:35 poems

troy@debian:~$ ls –l poems
total 8
-rw-r--rw-  1 troy troy 1030 Mar 20 21:35 jabberwocky
-rw-r-----  1 troy troy 3427 Mar 20 21:33 jumblies
```

Now, user `zapata` changes directory into `/home/troy` and enters the following sequence of commands also running a bash shell. Describe and explain the output obtained in each case. Remember, `zapata` is not a member of group troy, and assume that `jumblies` and `jabberwocky` are both non-empty plain text files.

i. `zapata@debian:/home/troy$ ls --al` **[3 Marks]**

ii. `zapata@debian:/home/troy$ ls –l poems` **[3 Marks]**

iii. `zapata@debian:/home/troy$ ls –l poems/jumblies` **[3 Marks]**

iv. `zapata@debian:/home/troy$ cat poems/jumblies` **[3 Marks]**

v. `zapata@debian:/home/troy$ cat poems/jabberwocky` **[3 Marks]**

End of Exam